Honors Calculus 1610

Lecture 16: Compositions, the Chain Rule, and Line Integrals

November 13, 2026

Today

- Open and closed balls; what "defined near $\vec{a}$" means.
- Limits of compositions, and continuity of compositions.
- The chain rule along a curve: $(f \circ \vec{r})'(t) = \nabla f(\vec{r}(t)) \cdot \vec{r}\,'(t)$.
- Directional derivatives.
- Vector fields.
- Line integrals, and the fundamental theorem for line integrals.

Balls

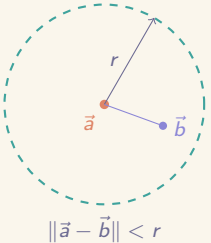
Definition

Let $\vec{a} \in \mathbb{R}^n$ and $r > 0$. The open ball of radius r around $\vec{a}$ is

$$B(\vec{a}, r) = \left\{ \vec{b} \in \mathbb{R}^n : \|\vec{a} - \vec{b}\| < r \right\},$$

and the closed ball is

$$\overline{B}(\vec{a}, r) = \left\{ \vec{b} \in \mathbb{R}^n : \|\vec{a} - \vec{b}\| \leq r \right\}.$$



Colloquially: "$\vec{f}$ is defined near $\vec{a}$" means that $\vec{f}$ is defined on $B(\vec{a}, r)$ for some $r > 0$, so $\vec{f} \colon B(\vec{a}, r) \to \mathbb{R}^m$.

Limits of compositions

Let $\vec{g}$ take values in $\mathbb{R}^m$ and be defined near $\vec{a} \in \mathbb{R}^n$, and let $\vec{f}$ take values in $\mathbb{R}^p$:

$$\mathbb{R}^n \xrightarrow{\vec{g}} \mathbb{R}^m \xrightarrow{\vec{f}} \mathbb{R}^p, \qquad \vec{f} \circ \vec{g} \colon \mathbb{R}^n \to \mathbb{R}^p.$$

Lemma (limit of a composition)

Suppose $\lim_{\vec{x} \to \vec{a}} \vec{g}(\vec{x}) = \vec{L}$, and that $\vec{f}$ is defined near $\vec{L}$ and continuous at $\vec{L}$. Then

$$\lim_{\vec{x} \to \vec{a}} \vec{f}(\vec{g}(\vec{x})) = \vec{f}(\vec{L}) = \vec{f}\Big(\lim_{\vec{x} \to \vec{a}} \vec{g}(\vec{x})\Big).$$

Without some hypothesis at the point $\vec{L}$ itself, this can fail: the proof shows exactly where it is needed.

Proof

Write $\vec{M} = \vec{f}(\vec{L})$. We want: for every $\varepsilon > 0$ there is $\delta > 0$ with

$$0 < \|\vec{x} - \vec{a}\| < \delta \implies \|\vec{f}(\vec{g}(\vec{x})) - \vec{M}\| < \varepsilon.$$

Since $\lim_{\vec{y} \to \vec{L}} \vec{f}(\vec{y}) = \vec{M}$, there is $\eta > 0$ with

$$0 < \|\vec{y} - \vec{L}\| < \eta \implies \|\vec{f}(\vec{y}) - \vec{M}\| < \varepsilon.$$

Since $\lim_{\vec{x} \to \vec{a}} \vec{g}(\vec{x}) = \vec{L}$, there is $\delta > 0$ with

$$0 < \|\vec{x} - \vec{a}\| < \delta \implies \|\vec{g}(\vec{x}) - \vec{L}\| < \eta.$$

Proof, continued

Now let $0 < \|\vec{x} - \vec{a}\| < \delta$ and put $\vec{y} = \vec{g}(\vec{x})$, so that $\|\vec{y} - \vec{L}\| < \eta$.

Either the length $\|\vec{y} - \vec{L}\|$ is 0 or it is not.

- If $\|\vec{y} - \vec{L}\| = 0$, then $\vec{y} = \vec{L}$. Since $\vec{f}$ is continuous at $\vec{L}$, $\vec{M} = \vec{f}(\vec{L})$, and plugging in,
$$\|\vec{f}(\vec{y}) - \vec{M}\| = \|\vec{f}(\vec{L}) - \vec{f}(\vec{L})\| = 0 < \varepsilon.$$
- If $\|\vec{y} - \vec{L}\| \neq 0$, then $0 < \|\vec{y} - \vec{L}\| < \eta$, and the definition of the limit of $\vec{f}$ at $\vec{L}$ gives $\|\vec{f}(\vec{y}) - \vec{M}\| < \varepsilon$.

Either way $\|\vec{f}(\vec{g}(\vec{x})) - \vec{M}\| < \varepsilon$. □

Continuity is needed only in the first case: the limit of $\vec{f}$ at $\vec{L}$ says nothing about $\vec{f}(\vec{L})$.

Continuity of compositions

Corollary

Let $\vec{g}\colon \mathbb{R}^n \to \mathbb{R}^m$ and $\vec{f}\colon \mathbb{R}^m \to \mathbb{R}^p$. If $\vec{g}$ is continuous at $\vec{a}$ and $\vec{f}$ is continuous at $\vec{g}(\vec{a})$, then $\vec{f} \circ \vec{g}$ is continuous at $\vec{a}$.

Proof. By the lemma, with $\vec{L} = \vec{g}(\vec{a})$:

$$\lim_{\vec{x}\to\vec{a}} \vec{f}(\vec{g}(\vec{x})) = \vec{f}\Big(\lim_{\vec{x}\to\vec{a}} \vec{g}(\vec{x})\Big) = \vec{f}(\vec{g}(\vec{a})). \qquad \square$$

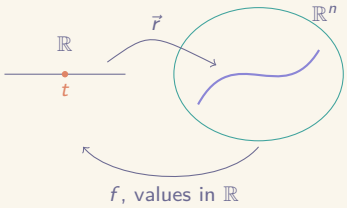
The chain rule along a curve

A curve $\vec{r}\colon \mathbb{R} \to \mathbb{R}^n$ and a function $f\colon \mathbb{R}^n \to \mathbb{R}$ give

$$f \circ \vec{r}\colon \mathbb{R} \to \mathbb{R},$$

e.g. the temperature along a path. We know $\vec{r}\,'(t)$, a vector-valued limit, and ∇f, the vector of partials.

The guess "$(f \circ \vec{r})' = f'(\vec{r})\,\vec{r}\,'$" becomes:



Theorem (chain rule)

If $\vec{r}$ is differentiable at t and f is differentiable at $\vec{r}(t)$, then

$$(f \circ \vec{r})'(t) = \nabla f(\vec{r}(t)) \cdot \vec{r}\,'(t).$$

The Carathéodory form

The nicest explanation uses the Carathéodory form of the derivative, as in one variable (Lecture 10). We state it without proof:

Fact

$f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\vec{b}$ if and only if there is a vector-valued $\vec{Q}$, continuous at $\vec{b}$, with

$$f(\vec{y}) - f(\vec{b}) = \vec{Q}(\vec{y}) \cdot (\vec{y} - \vec{b}) \qquad \text{for all } \vec{y},$$

and then $\vec{Q}(\vec{b}) = \nabla f(\vec{b})$.

In one variable, $Q(y)$ was the difference quotient $\dfrac{f(y) - f(b)}{y - b}$. In $\mathbb{R}^n$ we cannot divide by the vector $\vec{y} - \vec{b}$, so $\vec{Q}(\vec{y})$ is a vector playing the role of "the slope from $\vec{b}$ to $\vec{y}$".

The heuristic, and what $\vec{Q}$ is for

Imitating one variable, we would like to multiply and divide:

$$\frac{f(\vec{r}(t)) - f(\vec{r}(c))}{t - c} = \frac{f(\vec{r}(t)) - f(\vec{r}(c))}{\vec{r}(t) - \vec{r}(c)} \cdot \frac{\vec{r}(t) - \vec{r}(c)}{t - c}.$$

⚠ **Illegal: we cannot divide by the vector $\vec{r}(t) - \vec{r}(c)$.** ⚠

The Carathéodory form repairs this. With $\vec{b} = \vec{r}(c)$, $\vec{y} = \vec{r}(t)$,

$$f(\vec{y}) - f(\vec{b}) = \vec{Q}(\vec{y}) \cdot (\vec{y} - \vec{b}), \qquad \text{so} \qquad \vec{Q}(\vec{y}) \text{ plays the role of } \text{“}\frac{f(\vec{y}) - f(\vec{b})}{\vec{y} - \vec{b}}\text{”}$$ ⚠

an honest vector which, dotted with $\vec{y} - \vec{b}$, gives back $f(\vec{y}) - f(\vec{b})$.

Proof of the chain rule

Let $\vec{b} = \vec{r}(c)$ and take $\vec{Q}$ for f at $\vec{b}$. Put $\vec{y} = \vec{r}(t)$:

$$\frac{f(\vec{r}(t)) - f(\vec{r}(c))}{t - c} = \vec{Q}(\vec{r}(t)) \cdot \frac{\vec{r}(t) - \vec{r}(c)}{t - c}.$$

($\vec{Q}(\vec{r}(t))$ stands in for the illegal fraction of the previous slide.) As $t \to c$:

- $\vec{r}(t) \to \vec{r}(c)$ and $\vec{Q}$ is continuous at $\vec{r}(c)$, so $\vec{Q}(\vec{r}(t)) \to \vec{Q}(\vec{r}(c)) = \nabla f(\vec{r}(c))$ (composition lemma);
- $\dfrac{\vec{r}(t) - \vec{r}(c)}{t - c} \to \vec{r}\,'(c)$.

Limits are componentwise, so the dot product converges to $\nabla f(\vec{r}(c)) \cdot \vec{r}\,'(c)$. □

Directional derivatives

$\nabla f(\vec{a}) \cdot \vec{v}$ is the rate of change of f in the direction $\vec{v}$. Precisely:

Definition

For a unit vector $\vec{u}$, the directional derivative of f at $\vec{a}$ in the direction $\vec{u}$ is

$$D_{\vec{u}} f(\vec{a}) = \lim_{h \to 0} \frac{f(\vec{a} + h\vec{u}) - f(\vec{a})}{h}.$$

The partial derivatives are $D_{\vec{e}_i} f$.

Corollary

If f is differentiable at $\vec{a}$, then $D_{\vec{u}} f(\vec{a}) = \nabla f(\vec{a}) \cdot \vec{u}$.

Proof. The chain rule along $\vec{r}(t) = \vec{a} + t\vec{u}$, where $\vec{r}\,' = \vec{u}$. □

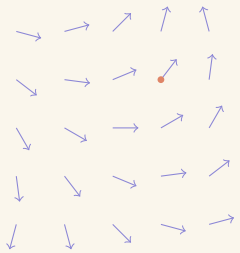
Vector fields

A function $\vec{F}\colon \mathbb{R}^n \to \mathbb{R}^n$ can be read in a new way:

- the input $\vec{x}$ is a point of space;
- the output $\vec{F}(\vec{x})$ is a vector attached at that point.

Read this way, $\vec{F}$ is a vector field: a force, a wind velocity, a flow, with an arrow at every point.

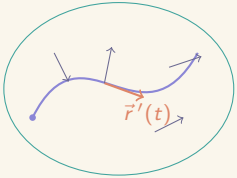
Example: $\vec{F} = \nabla G$ for $G\colon \mathbb{R}^n \to \mathbb{R}$.



Line integrals

A particle moves along $\vec{r}\colon [a,b] \to \mathbb{R}^n$ through a force field $\vec{F}\colon \mathbb{R}^n \to \mathbb{R}^n$. At time t:

- $\vec{r}\,'(t)$ is the velocity, a vector in $\mathbb{R}^n$;
- $\vec{F}(\vec{r}(t))$ is the force there, a vector in $\mathbb{R}^n$;
- $\vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t)$ is the infinitesimal work done at time t: to what extent the motion is lined up with the field.



Line integrals: the definition

Adding up the infinitesimal work over $[a, b]$:

Definition

If $\vec{F}$ is continuous and $\vec{r}\,'$ is continuous on $[a, b]$, the line integral of $\vec{F}$ along the curve C traced by $\vec{r}$ is

$$\int_C \vec{F} \cdot d\vec{r} = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t)\, dt,$$

the integral of a real-valued function of t.

The integrand is continuous (compositions and dot products of continuous functions), so the integral exists.

The fundamental theorem for line integrals

Theorem

Let $G\colon \mathbb{R}^n \to \mathbb{R}$ be differentiable with ∇G continuous, and let $\vec{r}\colon [a,b] \to \mathbb{R}^n$ have continuous derivative. Then

$$\int_C \nabla G \cdot d\vec{r} = G(\vec{r}(b)) - G(\vec{r}(a)).$$

Proof. By the chain rule, $\nabla G(\vec{r}(t)) \cdot \vec{r}\,'(t) = (G \circ \vec{r})'(t)$, which is continuous, so by the fundamental theorem of calculus

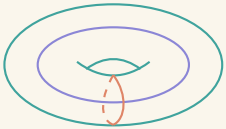
$$\int_a^b \nabla G(\vec{r}(t)) \cdot \vec{r}\,'(t)\,dt = \int_a^b (G \circ \vec{r})'(t)\,dt = G(\vec{r}(b)) - G(\vec{r}(a)). \qquad \square$$

For a gradient field, the work depends only on the endpoints.

An aside: loops

If $\vec{r}(a) = \vec{r}(b)$, the curve is a closed loop, and a gradient field does no work around it:

$$\int_C \nabla G \cdot d\vec{r} = 0.$$



Which fields are gradients, and what the integrals around loops detect about the shape of the space, are deep questions. They go by names like monodromy, holonomy, homology and de Rham cohomology.